

Supraspinal motor systems: cerebral cortex and brainstem

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Session Objectives

- 1) Understand the origin and destination of descending cerebral motor tracts
- 2) Identify the organization and destination of descending brainstem tracts
- 3) Describe the organization and capabilities of the corticospinal tract
- 4) Describe the organization and capabilities of the corticobulbar tract
- 5) Understand the difference between upper and lower motor neurons and their associated dysfunctions as well as brainstem damage

Nolte's The Human Brain: An introduction to its functional anatomy

Chapter 18 Overview of motor systems

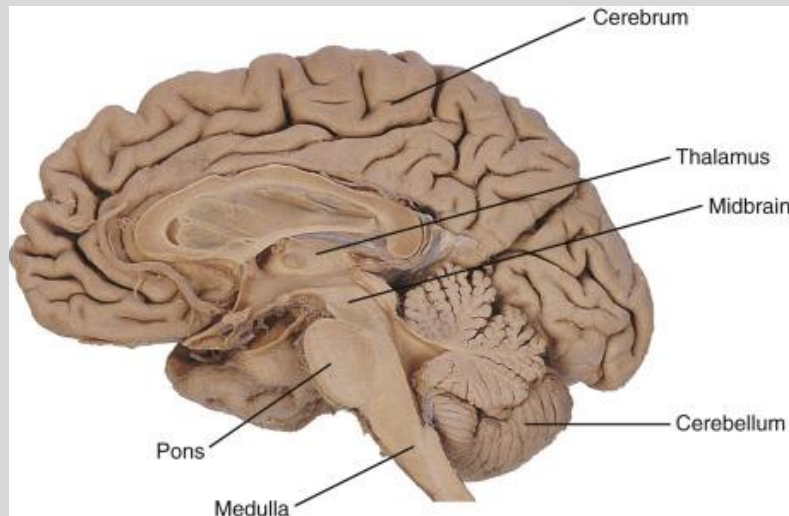
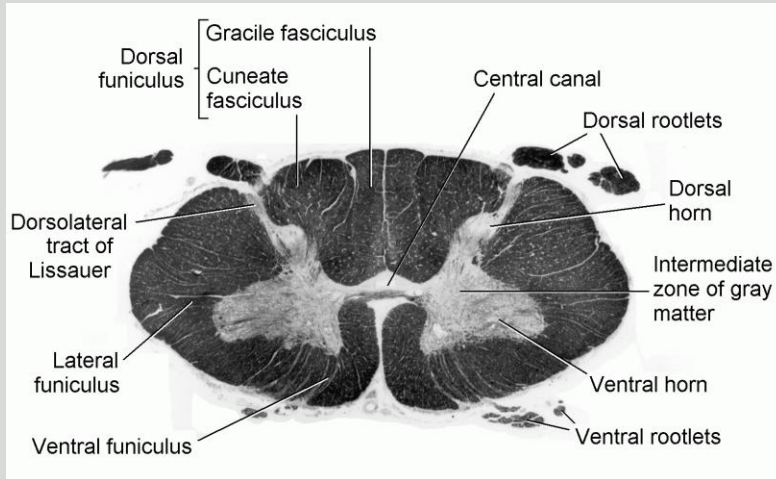
"Motor Control Systems Involve Both Hierarchical and Parallel Connections" and afterwards.

<http://arktos.nyit.edu/login?url=https://www-clinicalkey-com.arktos.nyit.edu/#!/content/book/3-s2.0-B9780323653985000187>

Lower motor neurons versus Upper motor neurons

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Nerves and ganglia are in PNS.
Tracts and nuclei are in CNS.

Lower motor neurons (also termed “motoneurons”) innervate skeletal muscle.

Cells bodies in spinal cord or cranial nerve nuclei.
Their tracts rarely decussate.
Their nerves never decussate.

Upper motor neurons have cell bodies at the cerebral cortex and brainstem. These project to lower motor neurons of the brainstem and spinal cord, not to muscles.

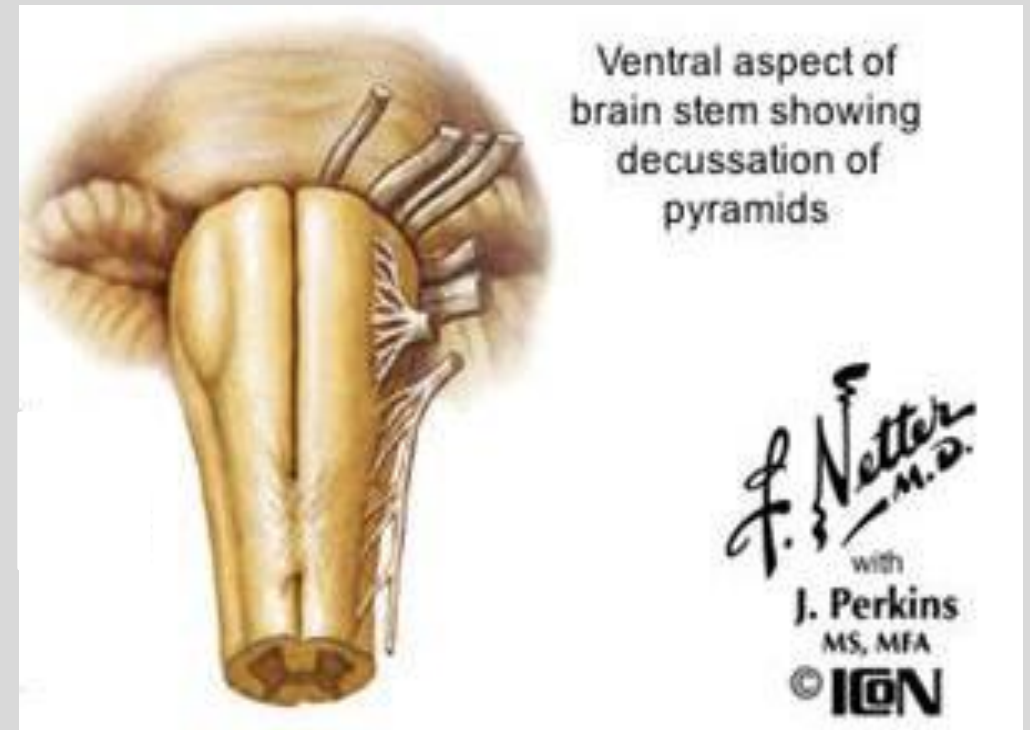
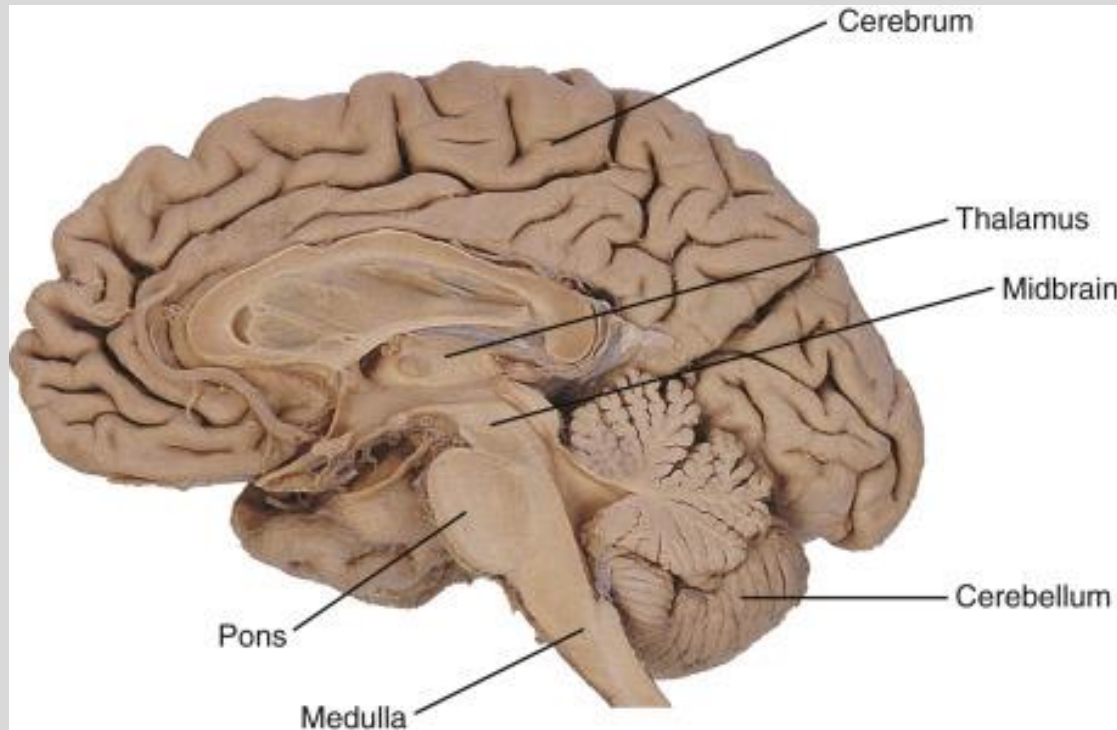
Variety of projection patterns: some decussate, some don't, and some project bilaterally.

The pyramidal system = cerebral cortical projections to the **spinal cord** and **brainstem**

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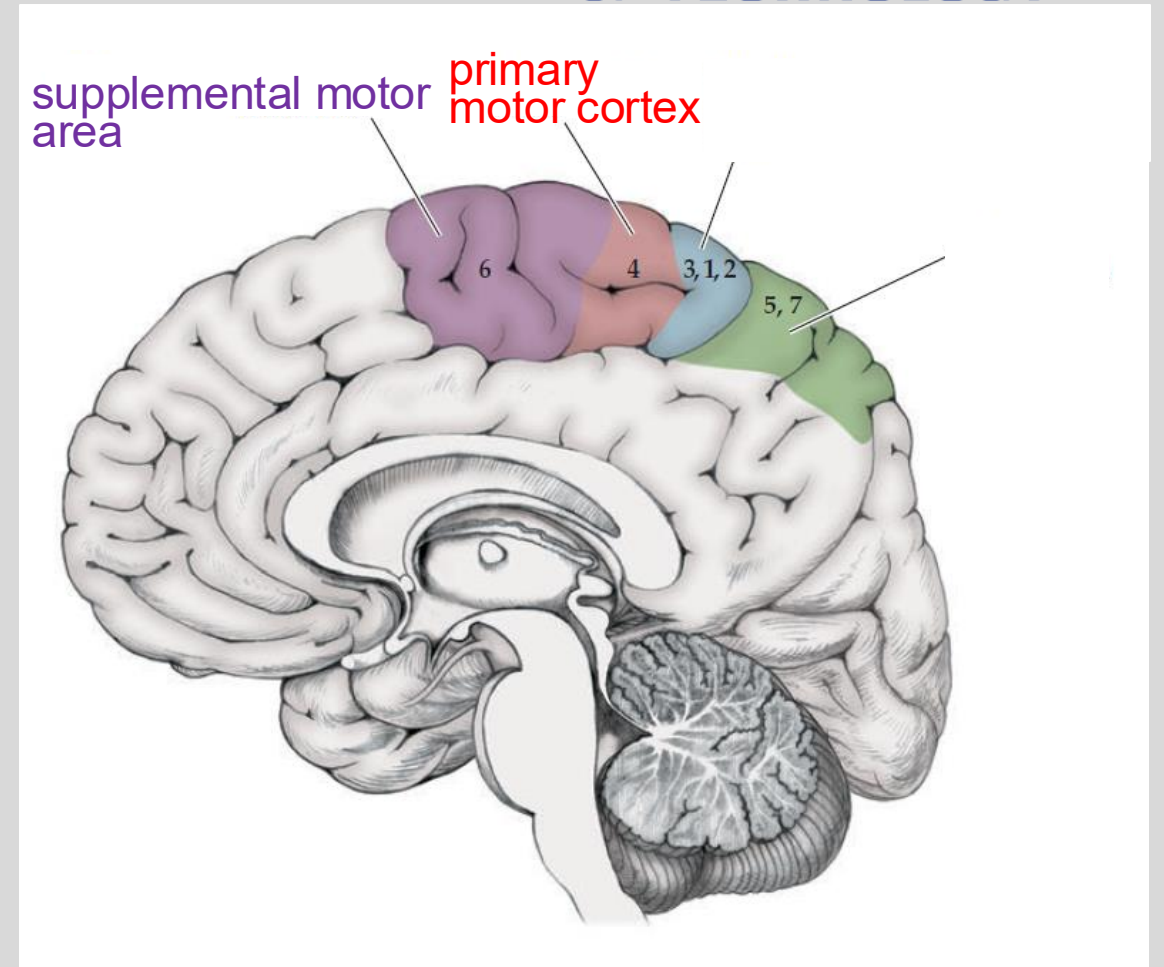
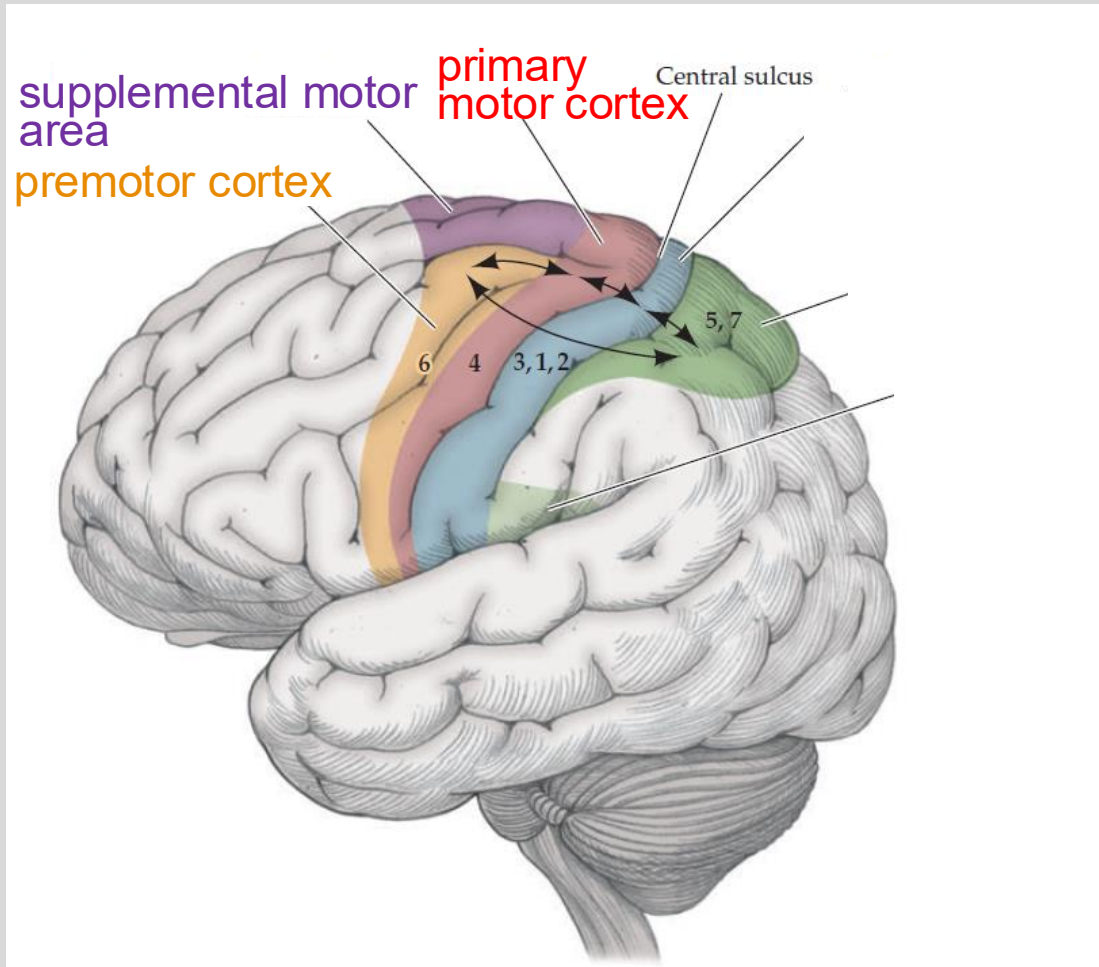
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Confusing designation since only the **corticospinal tract** (for the body and limbs) forms the medullary pyramids, not the **corticobulbar tracts** (for the cranial nerve nuclei).



Pyramidal system involves several frontal areas

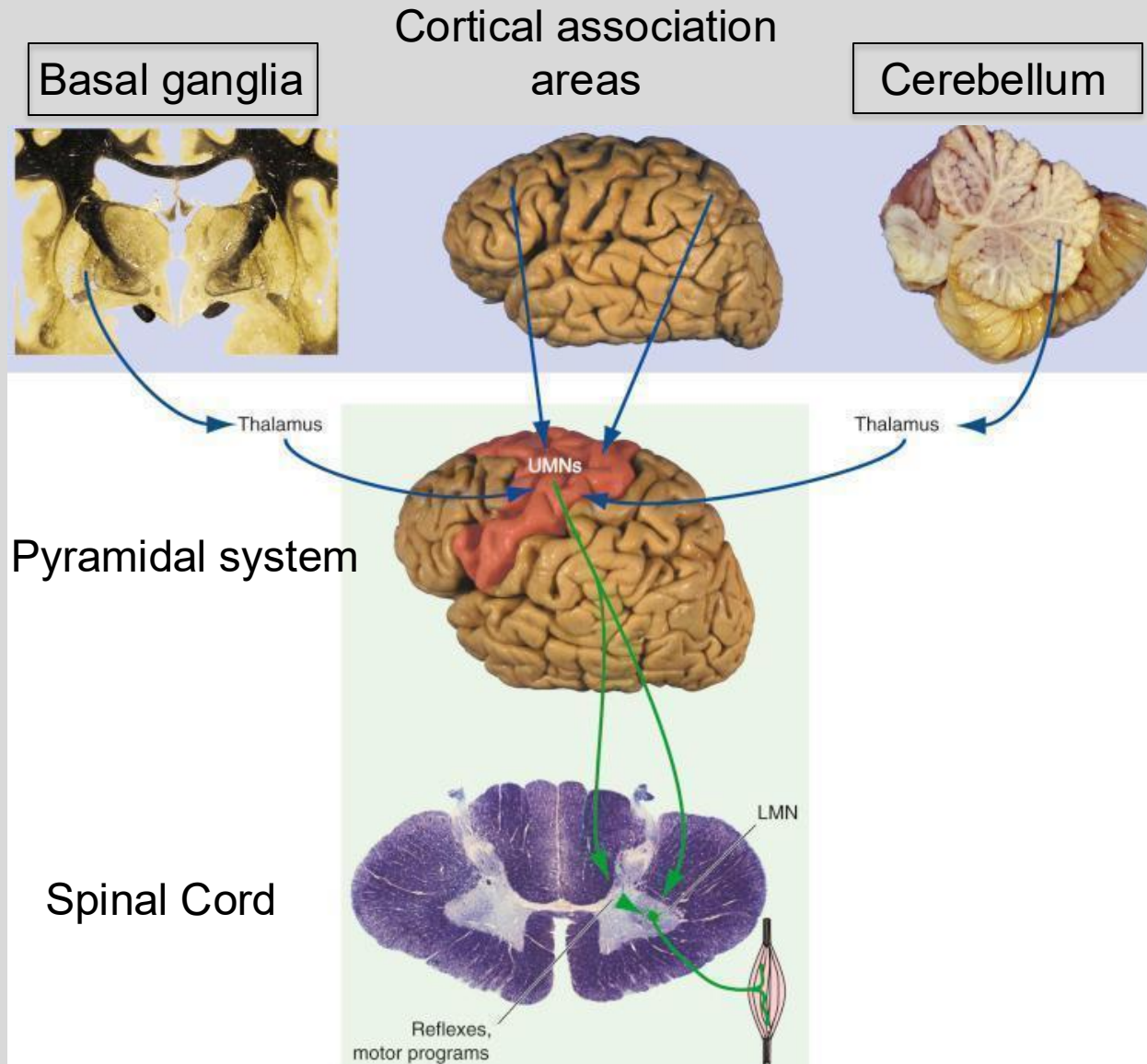
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Converging inputs to the pyramidal system

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These systems do not act on lower motor neurons, but on upper motor neurons of the cerebral cortex and brainstem.

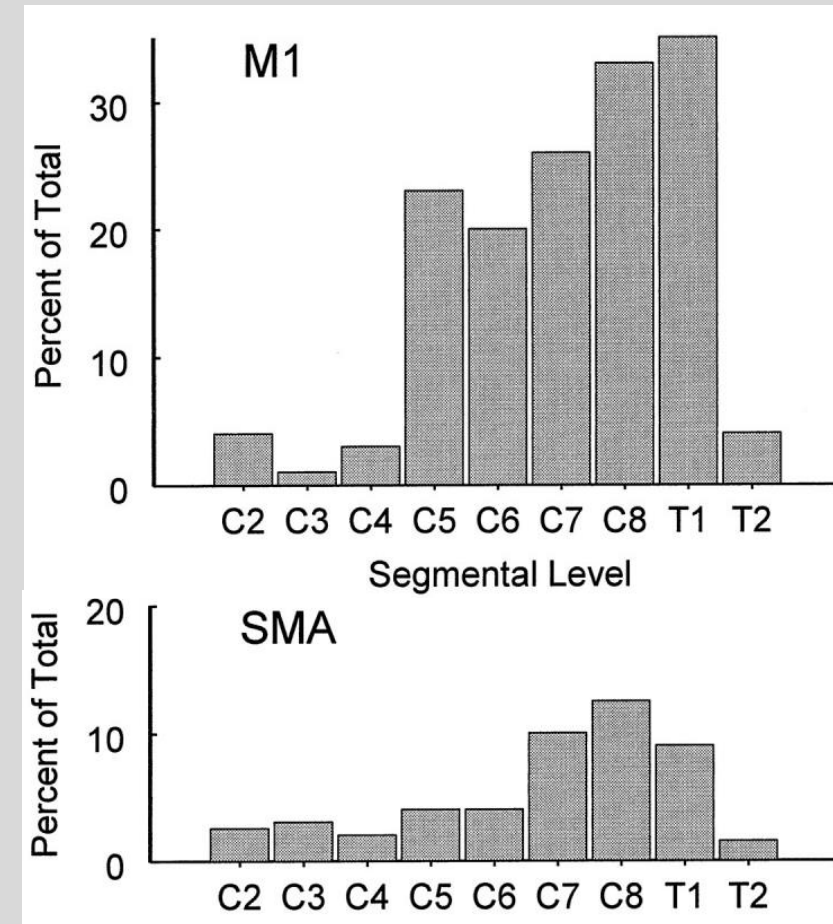
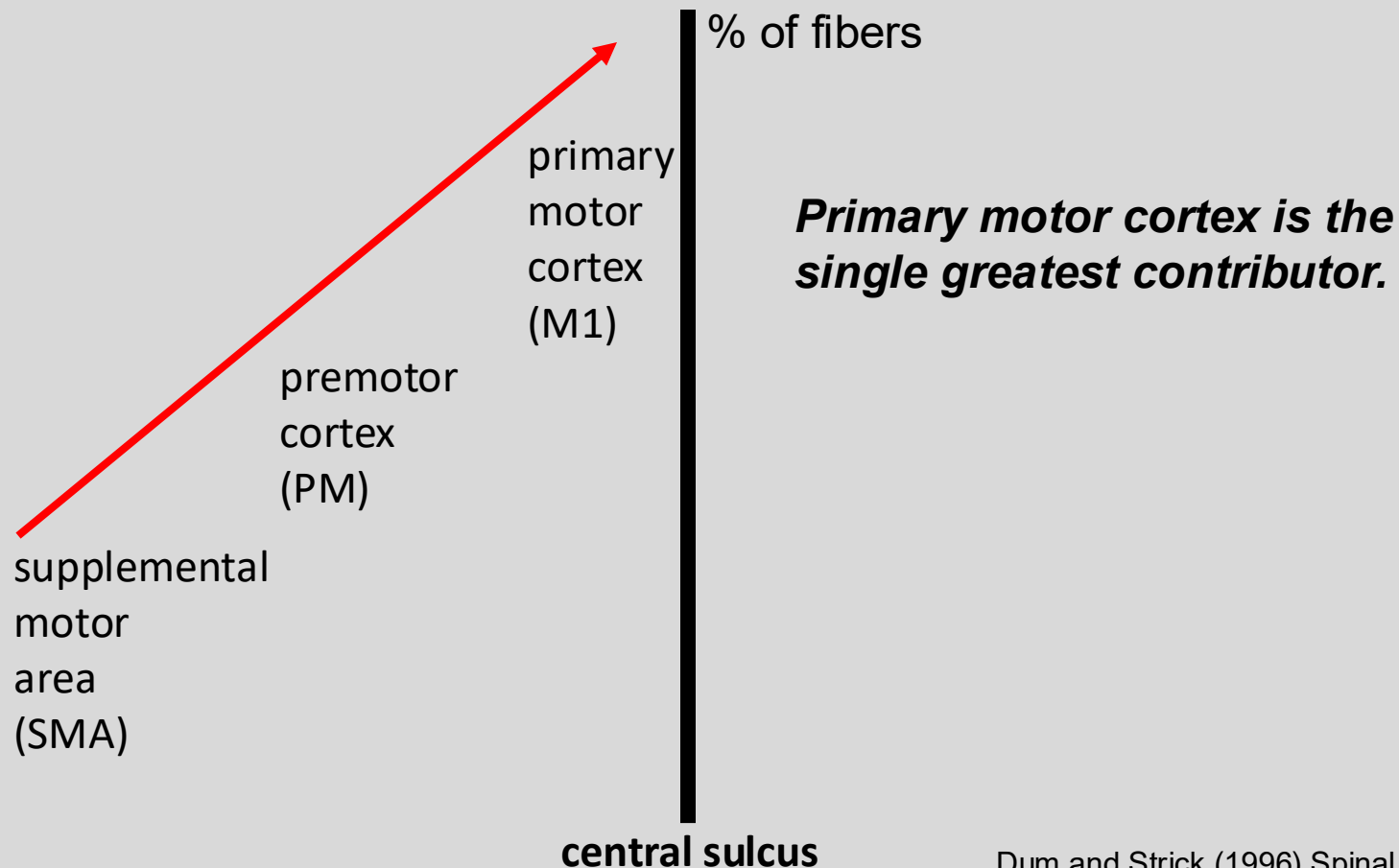
That's why the basal ganglia and cerebellum are called “extra-pyramidal”.

“Extrapyramidal symptoms” are dysfunctions of the basal ganglia (typically) and cerebellum.

Cerebral cortex (e.g. pyramidal system) is linked to ipsilateral basal ganglia and contralateral cerebellum.

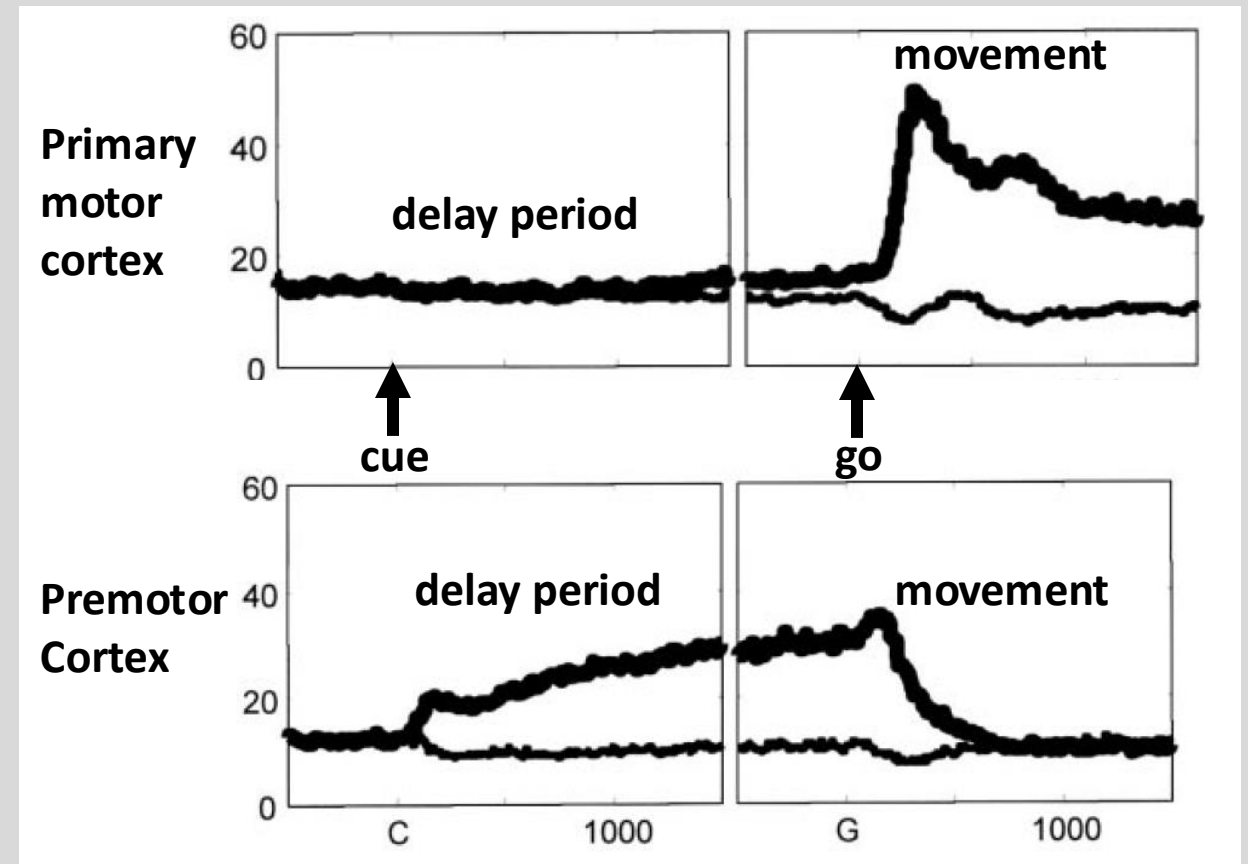
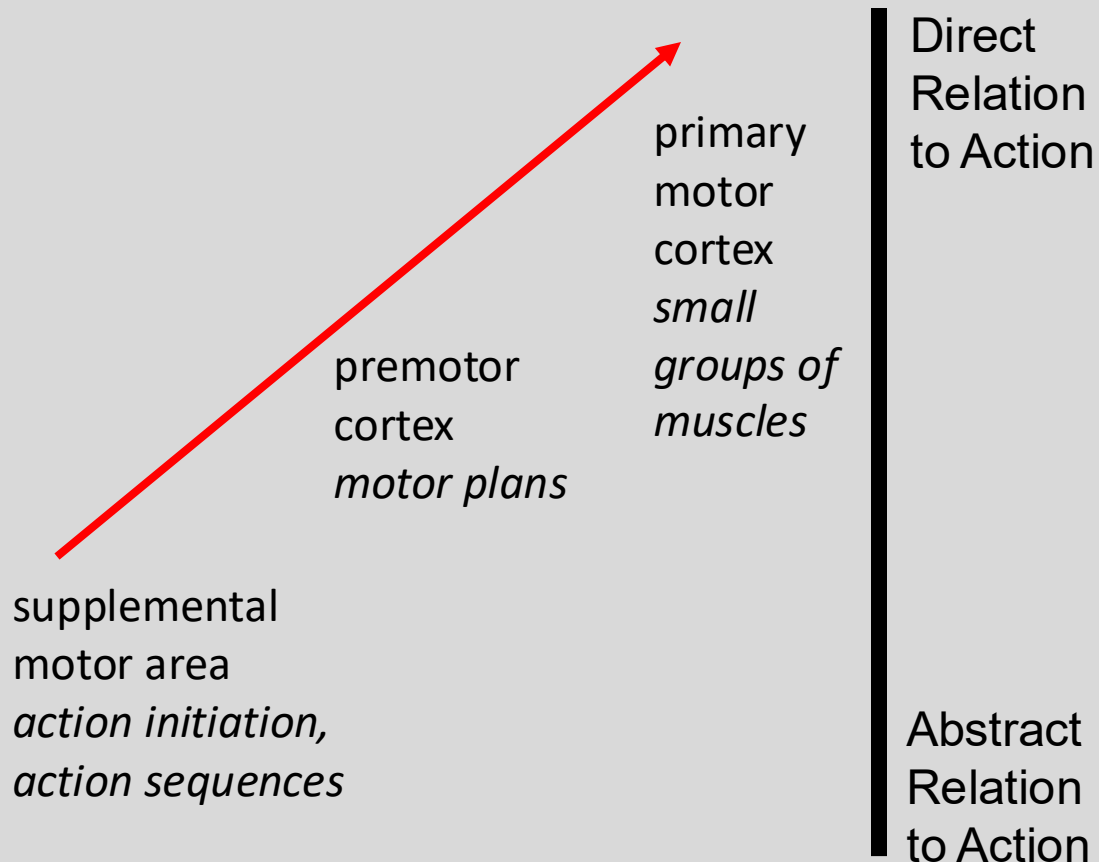
Regional specialization of pyramidal system

Regions closer to the central sulcus contribute more fibers



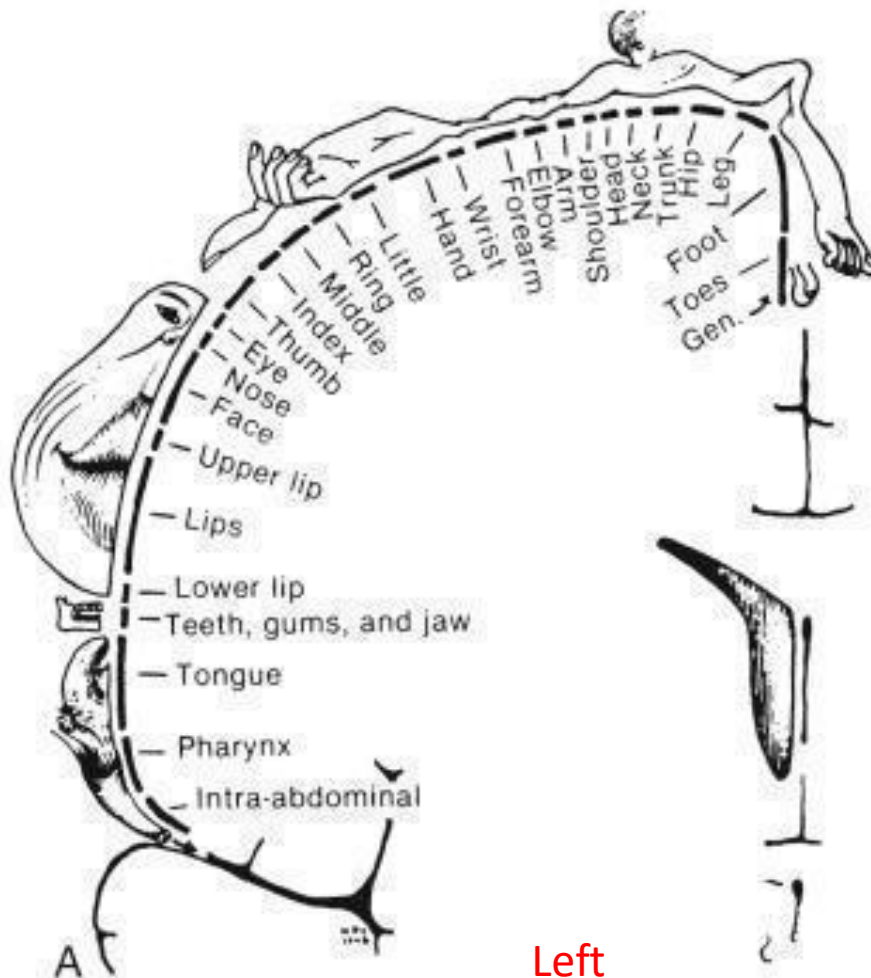
Regional specialization of pyramidal system

Regions closer to the central sulcus are more directly related to muscle activity



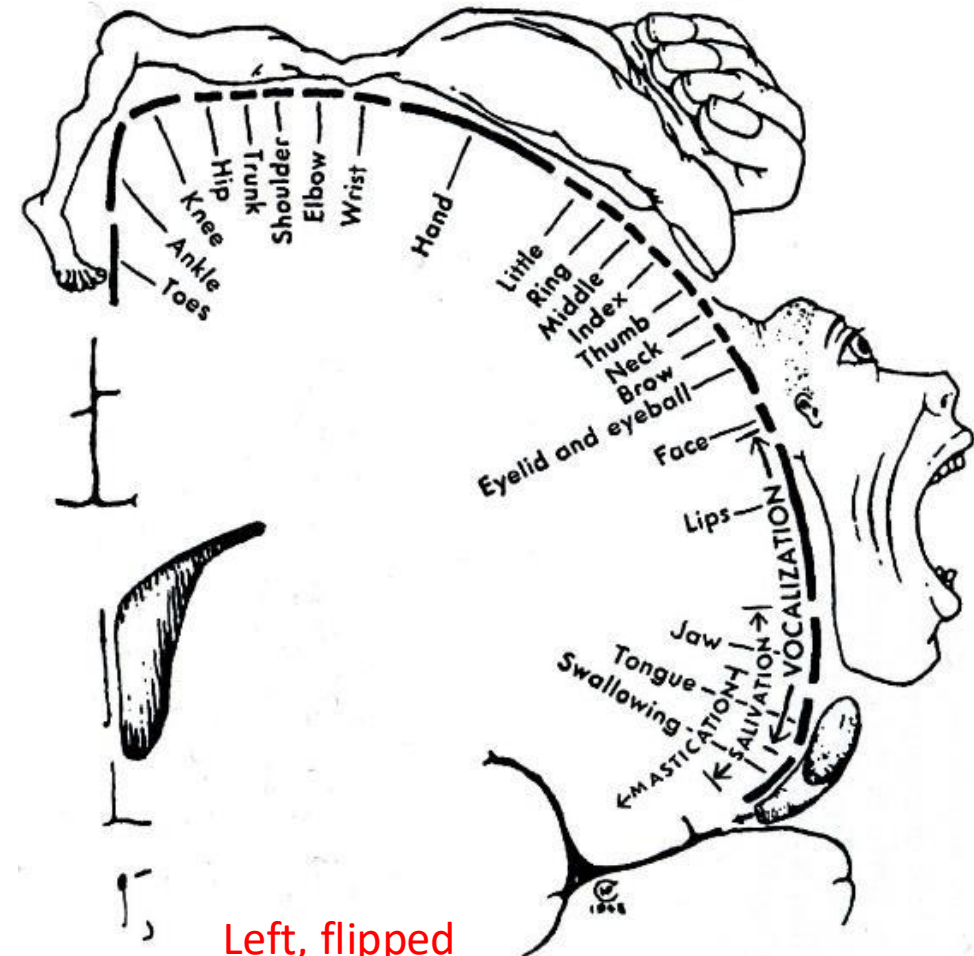
Similar layout of somatosensory and motor representations within cerebral cortex

Sensory homunculus (S1)



Left

Motor homunculus (M1)



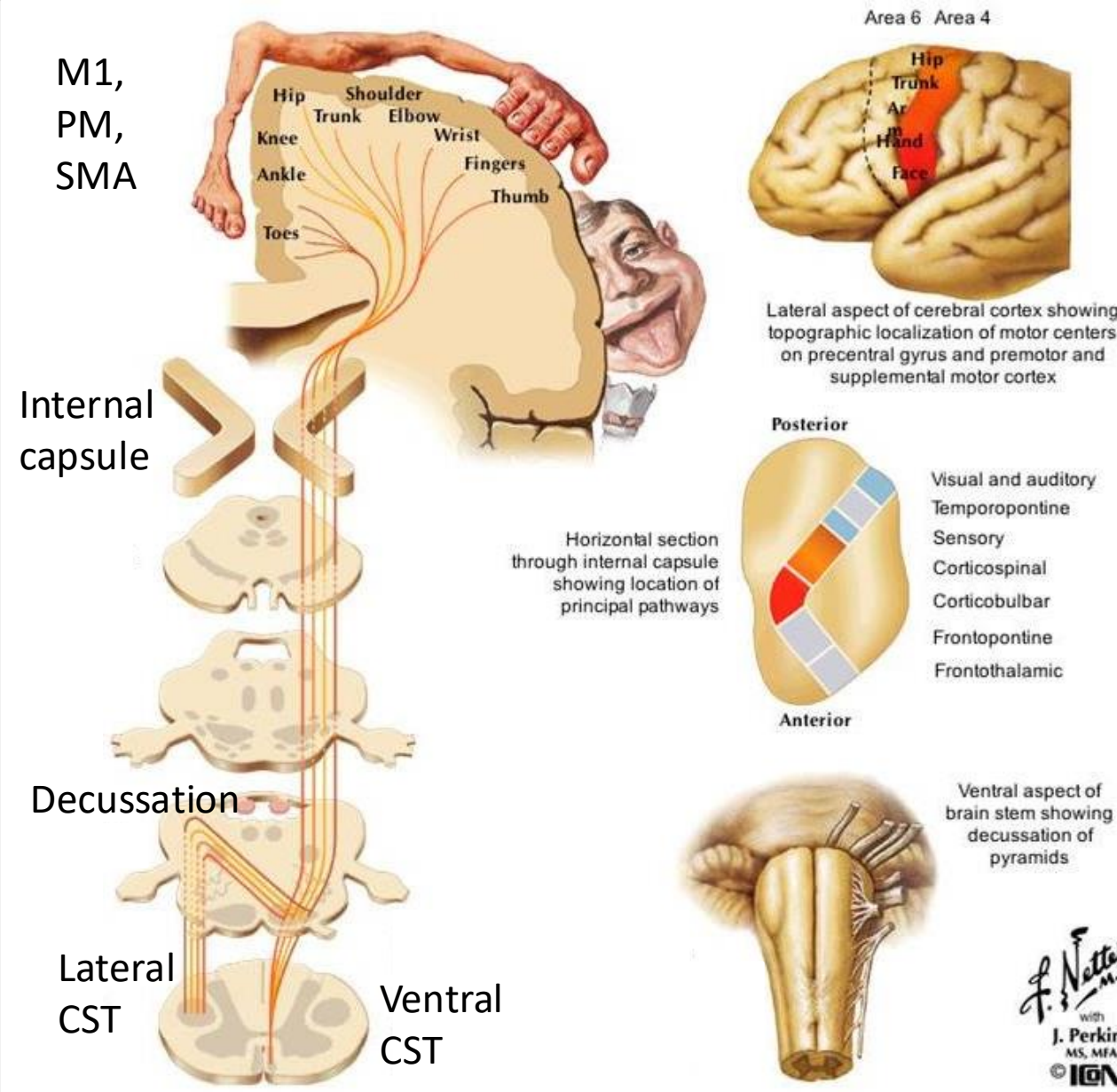
Left, flipped

central
sulcus

The two paths of the corticospinal tract

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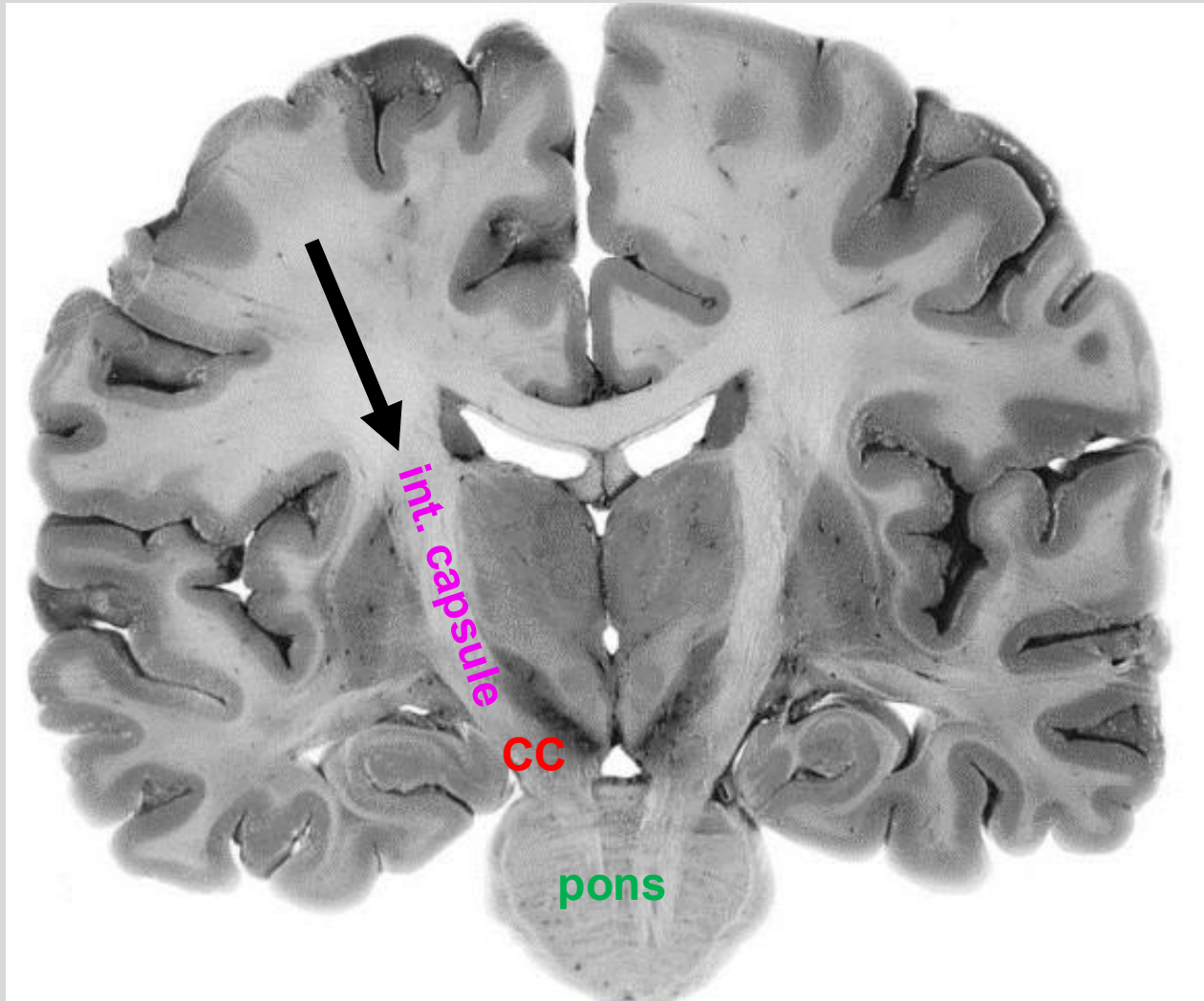
“Body-representation” fibers descend from cerebral cortex.

At the lower medulla, 80-85% fibers decussate and form the **lateral corticospinal tract**.

15-20% of fibers do not cross and descend ipsilaterally as the **ventral corticospinal tract**. Most of these decussate in the spinal cord.

Thereby, cerebral cortex controls the contralateral body

Corticospinal tract



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Corticospinal fibers converge and pass downward through the posterior limb of the **internal capsule***, continue as the **crus cerebri**, disperse in the ventral **pons**, regroup in the medulla and come to the ventral surface as the pyramid where they mostly decussate at the lowermost portion and descend the spinal cord to their relevant level.

Damage to anywhere above decussation leads to motor deficits on the contralateral side of the body.

Damage to the lateral spinal cord leads to motor deficits on the same side.

Lesion of corticospinal tract (upper motor neuron)

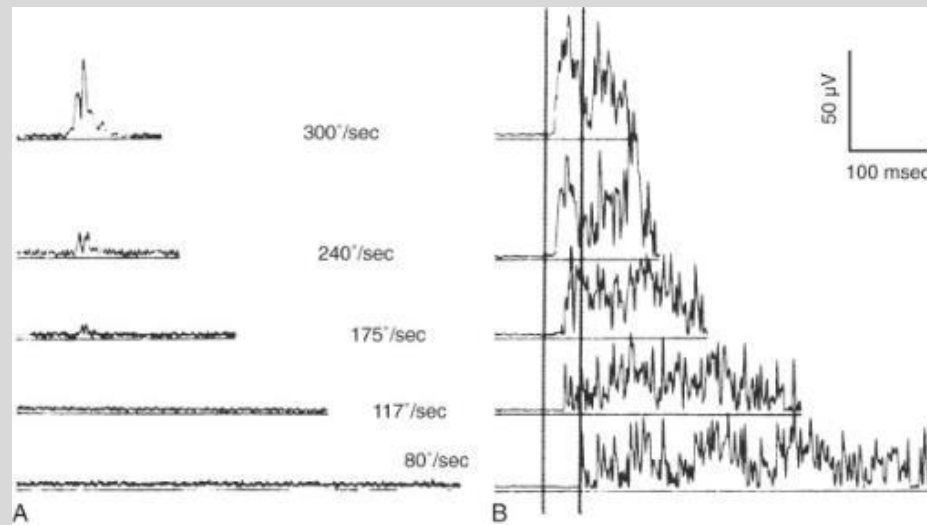
- Stroke, spinal cord injury, multiple sclerosis, cerebral palsy
- Impaired voluntary movement
- Diminished strength, speed, and agility
- Proximal joints and axial movements are less severely affected
- Positive clinical symptoms reflect the “release” of brainstem and spinal circuits previously dormant or well-managed.
- Increased stretch reflexes, abnormal synergies, & babinski sign

Lesion of corticospinal tract (upper motor neurons)

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Augmented stretch reflexes



Healthy

Clinical

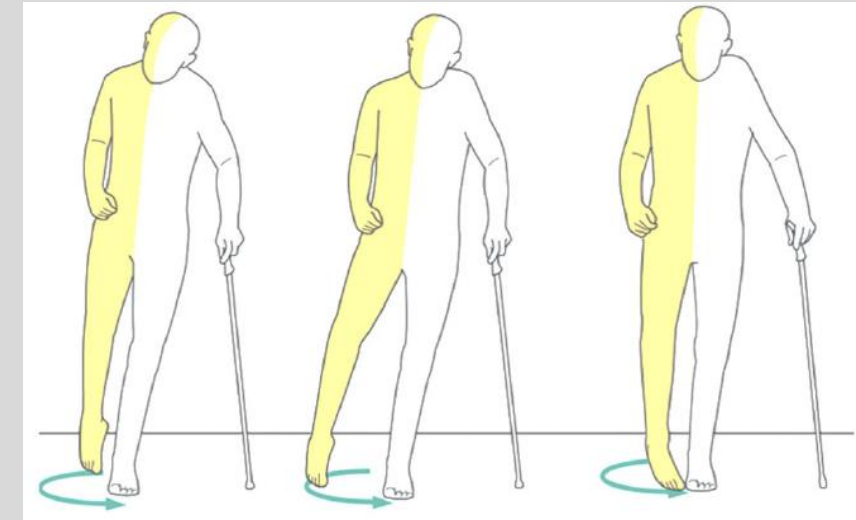
Thillman et al (1991) The mechanism of spastic muscle hypertonus.
Variation in reflex gain over the time course of spasticity.

Abnormal “synergies”



Involuntary elbow and wrist flexion especially during arm abduction

Altered walking



“Circumduction gait” - compensates for poor weight-bearing and ability to lift leg

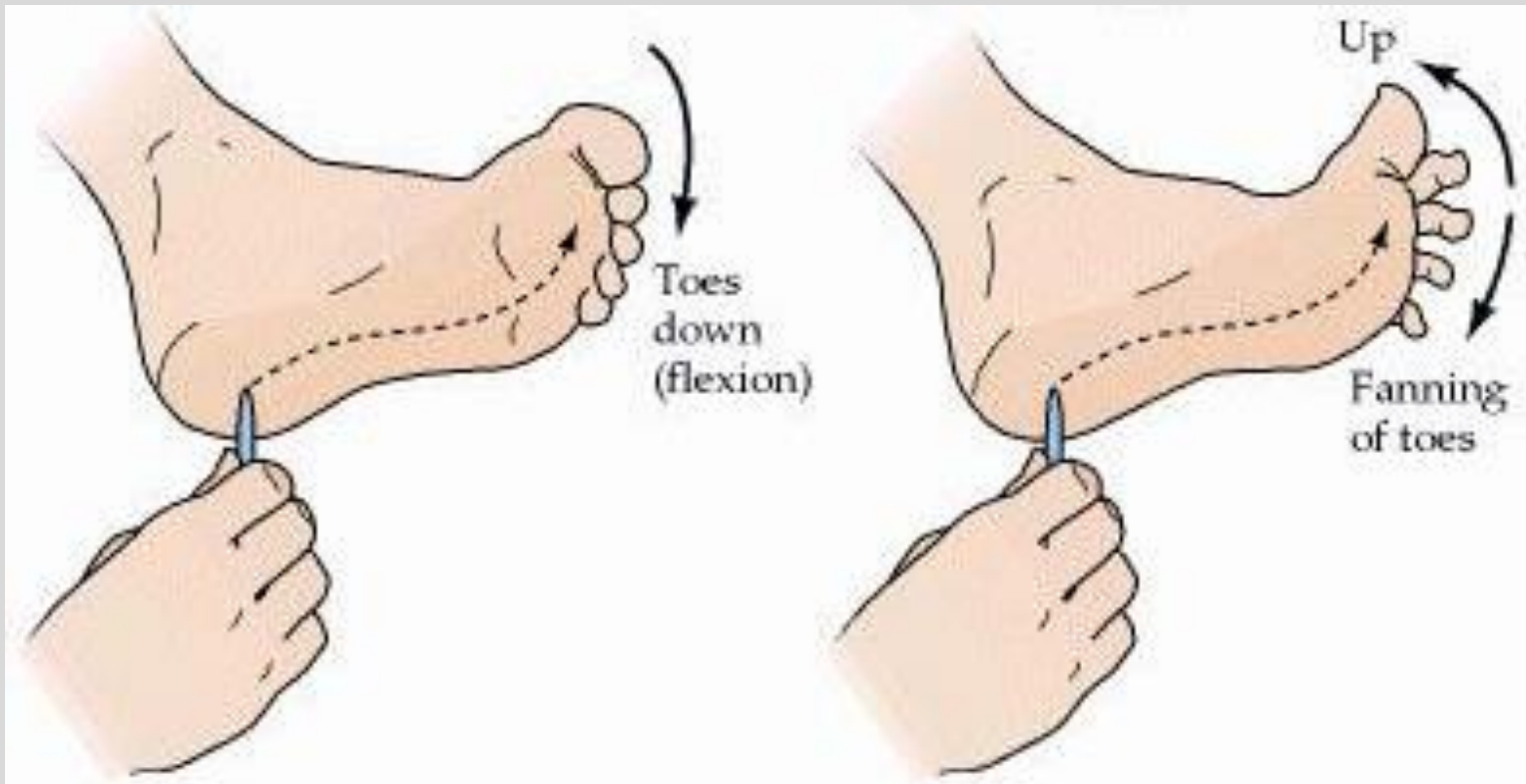
<https://www.youtube.com/watch?v=y160w4sAQNw>

Spasticity from stroke or spinal cord damage can take days to weeks to develop.
The immediate motor symptom is weakness and flaccidity called “spinal shock”.

Lesion of corticospinal tract (upper motor neurons)

Normal

Babinski sign



Normally present for infants < 1 year

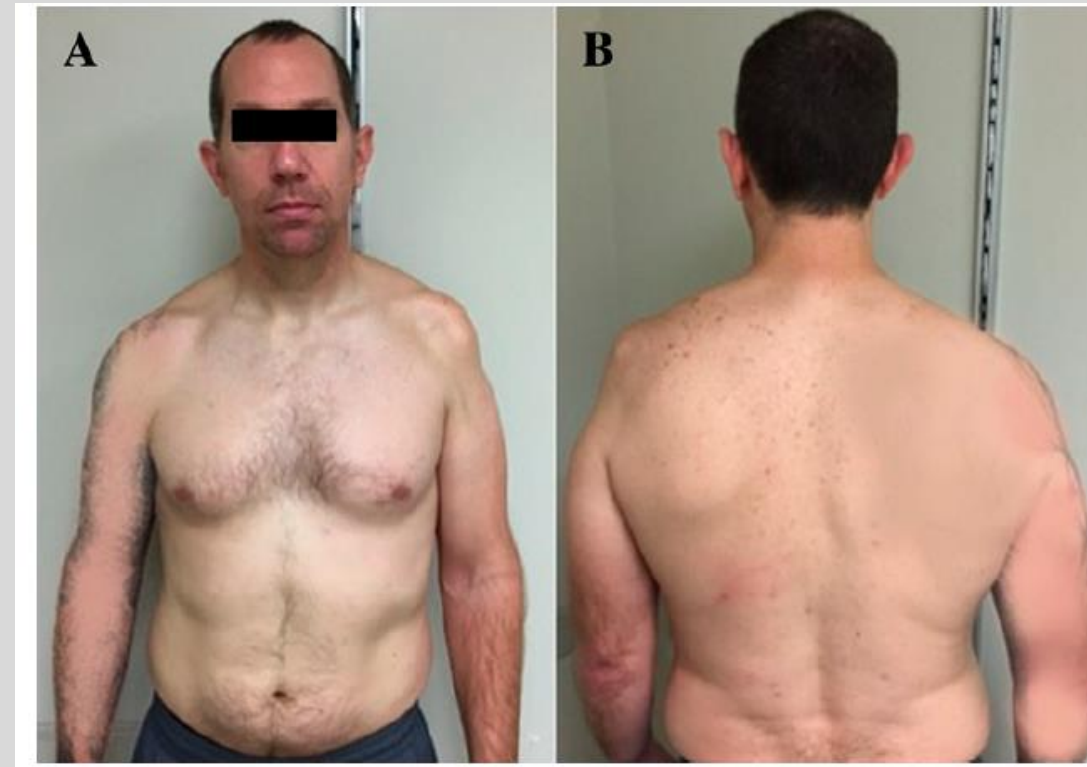
Also post-seizure states via
diminished/altered pyramidal control

Lesion of ventral horn (lower motor neuron) or its efferent nerve

Flaccid paralysis (as opposed to spastic paralysis)

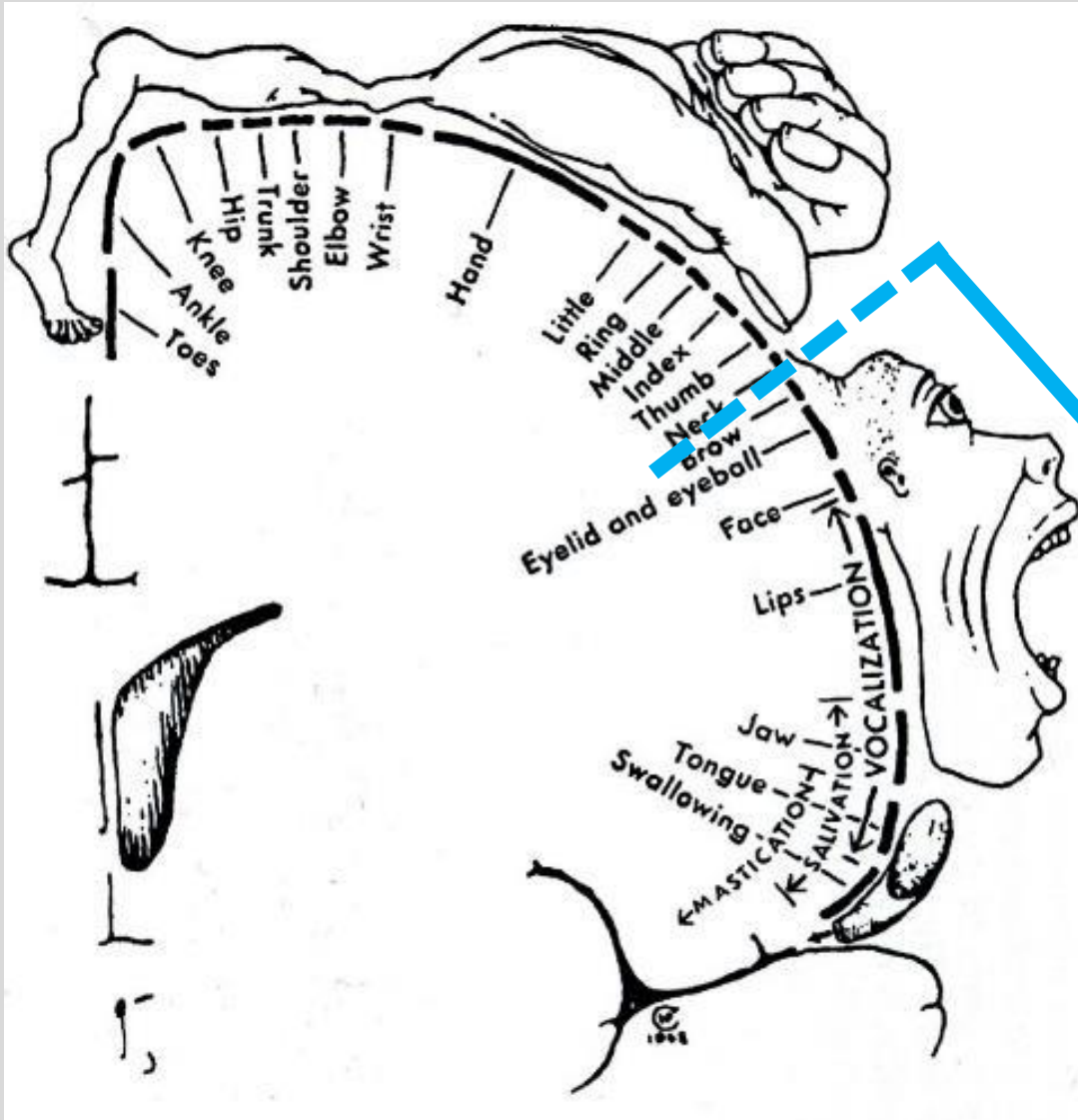
- Diminished voluntary strength
- Diminished reflexes & muscle tone
- Fasciculations (spontaneous muscle twitches)
- Muscle atrophy (takes time)

Example shows atrophy of left deltoid due to damage of axillary nerve.



Eichinger (2016) Outcomes following closed axillary nerve injury

Corticobulbar tract



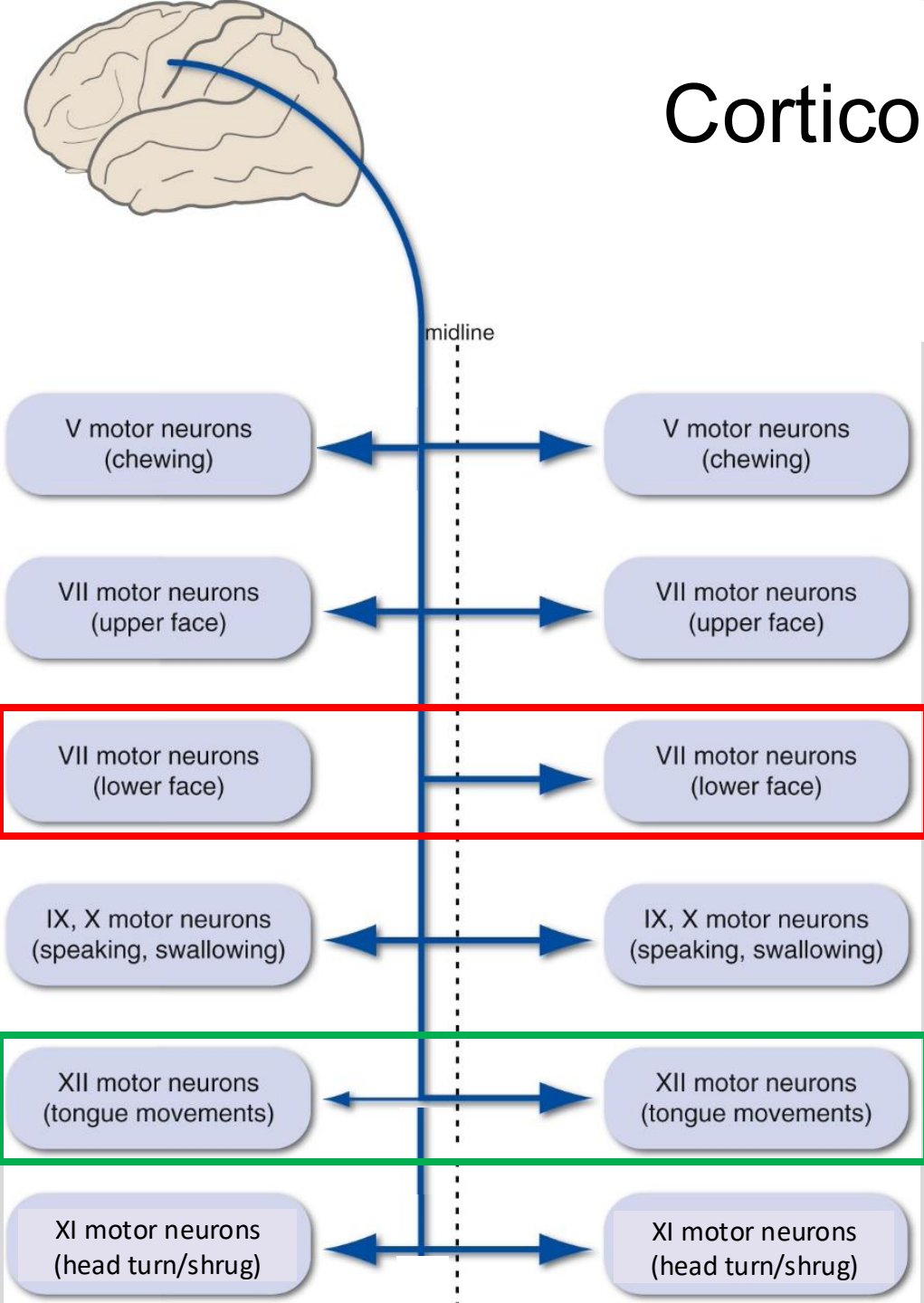
More ventral origin than corticospinal tract.

Projects to cranial nerve nuclei in the brain stem.

Cerebral cortex = “upper motor neurons”

Cranial nerve nuclei = “lower motor neurons”

Corticobulbar tract



Bilaterally innervates cranial nuclei controlling jaw, tongue, larynx, pharynx, and scm/upper trapezius.

EXCEPT: *Fibers projecting to **facial nucleus** show distinctive pattern:*

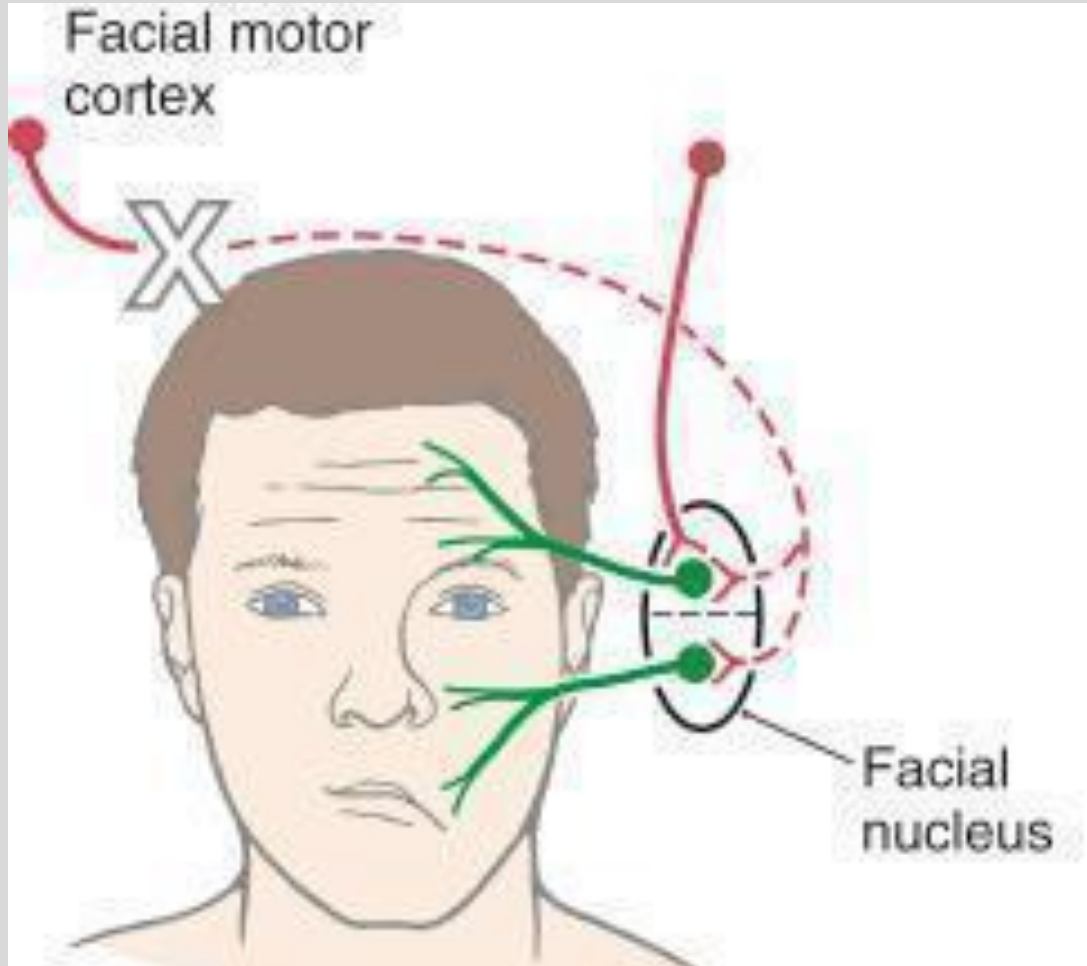
Contralateral nucleus receives innervation from whole corticobulbar tract (upper and lower face).

Ipsilateral facial nucleus only receives innervation for controlling upper face.

No redundancy for lower face. No compensation for loss.

Input to all other nuclei is bilateral though hypoglossal has more from contralateral cerebral cortex, i.e. partial compensation with tongue control.

Lesioning corticobulbar tract (upper motor neuron)



CN 7/Facial nerve

Remember forehead /eyebrow cortical motor neurons project **bilaterally** to facial motor nuclei, but cheek cortical motor neurons only project **contralaterally** to facial motor nuclei

Patient can raise their eyebrows and close eyelids, but cannot voluntarily move their lower 1/2 of face

Motor issues from a unilateral lesion of the corticobulbar track can also include the tongue -> deviation away from the lesioned side of cerebral cortex.

Lesioning corticobulbar tract (upper motor neuron)

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Lesion of cranial nerve or nuclei (lower motor neuron)

Flaccid paralysis: loss of muscle tone and voluntary movement of affected muscles, and atrophy of muscle

Bell's palsy

- Idiopathic paralysis of facial nerve (CN VII)
- Inability to control the affected side of face
- Sudden onset, often resolves without intervention
- Also issues with tearing, hearing, taste, and salivation



Lesion of cranial nerve or nuclei (lower motor neuron)

Hypoglossal nerve palsy

- Destruction of the hypoglossal nucleus or nerve
- Tongue deviates to the weak side due to unopposed action of contralateral genioglossus muscle

Determining whether UMN or LMN requires looking at other possible symptoms.

Facial control? What about the hand?

Limited cortical stroke could have similar features

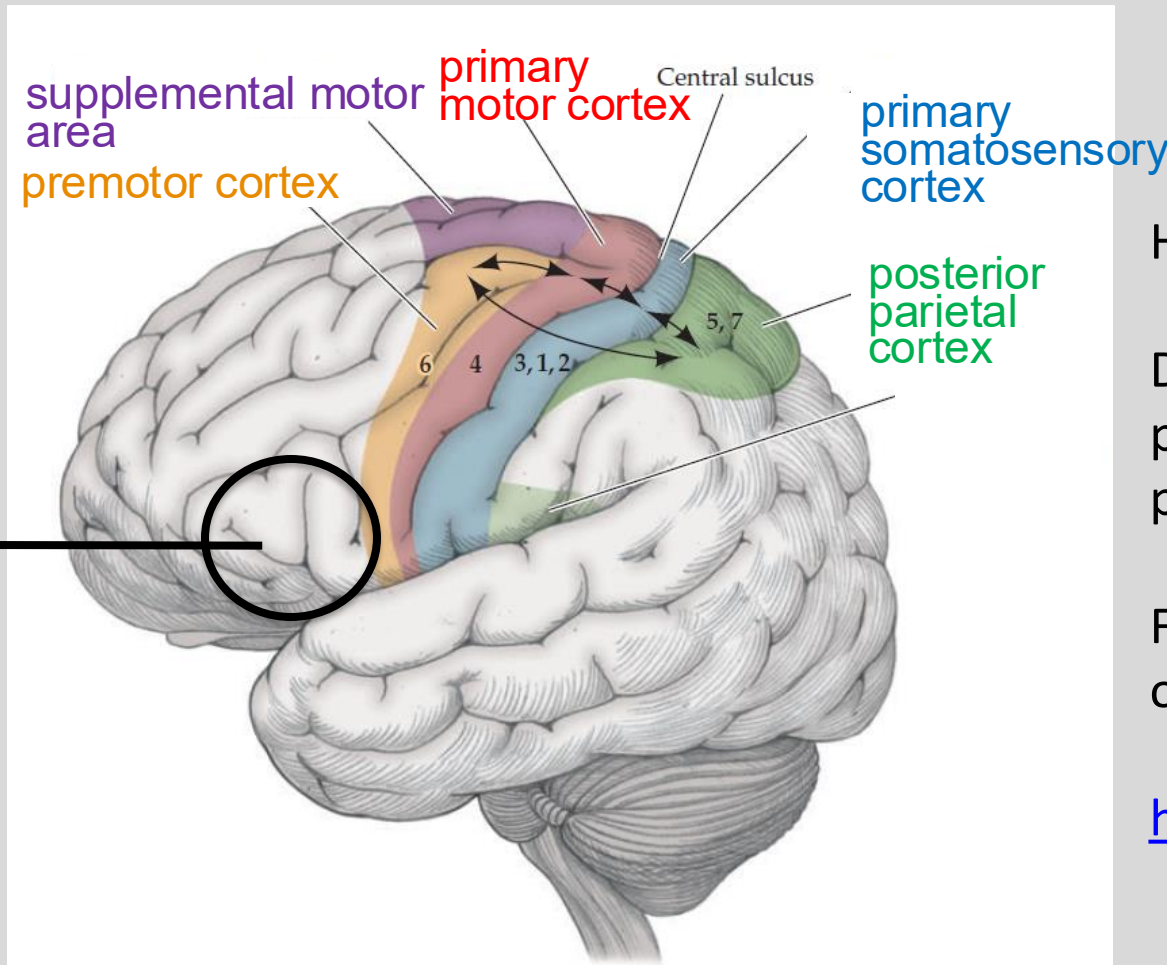


Cruz (2014) Unilateral lingual paralysis after isolated unilateral infarction in the primary motor cortex

Broca's area is spatially aligned with the facial homunculus

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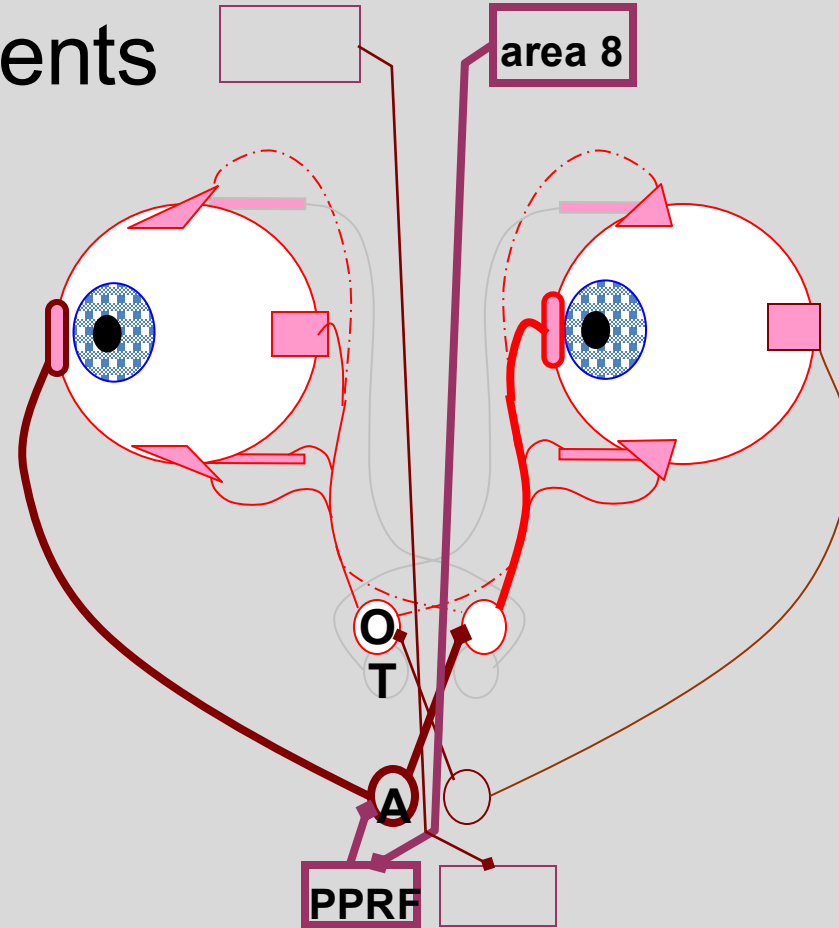
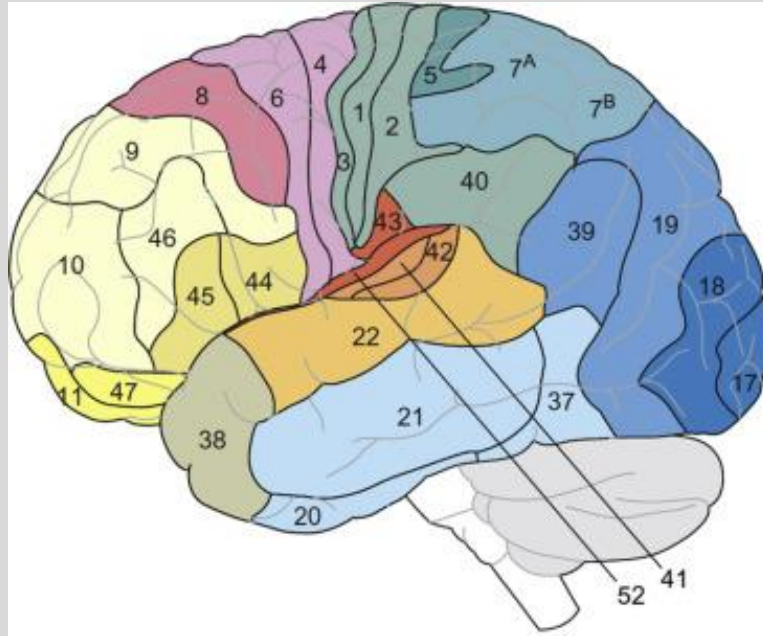
High order motor control specific for language.

Damage leads to 'expressive aphasia'. Inability to produce fluid spontaneous speech. May be able to produce some words, but few and at great effort.

Facial expressions, laughter, chewing, mouth opening, and swallowing are unaffected!

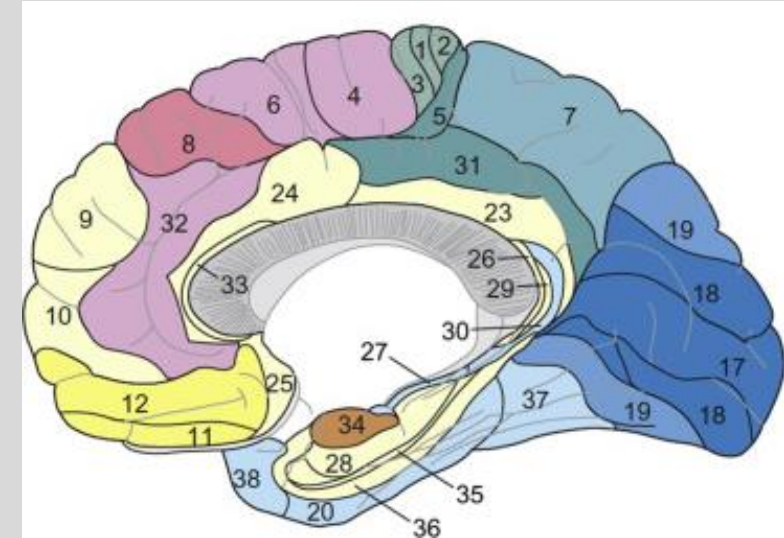
<https://www.youtube.com/watch?v=RMa9BVpJkYQ>

“Frontal eye fields” control voluntary eye movements

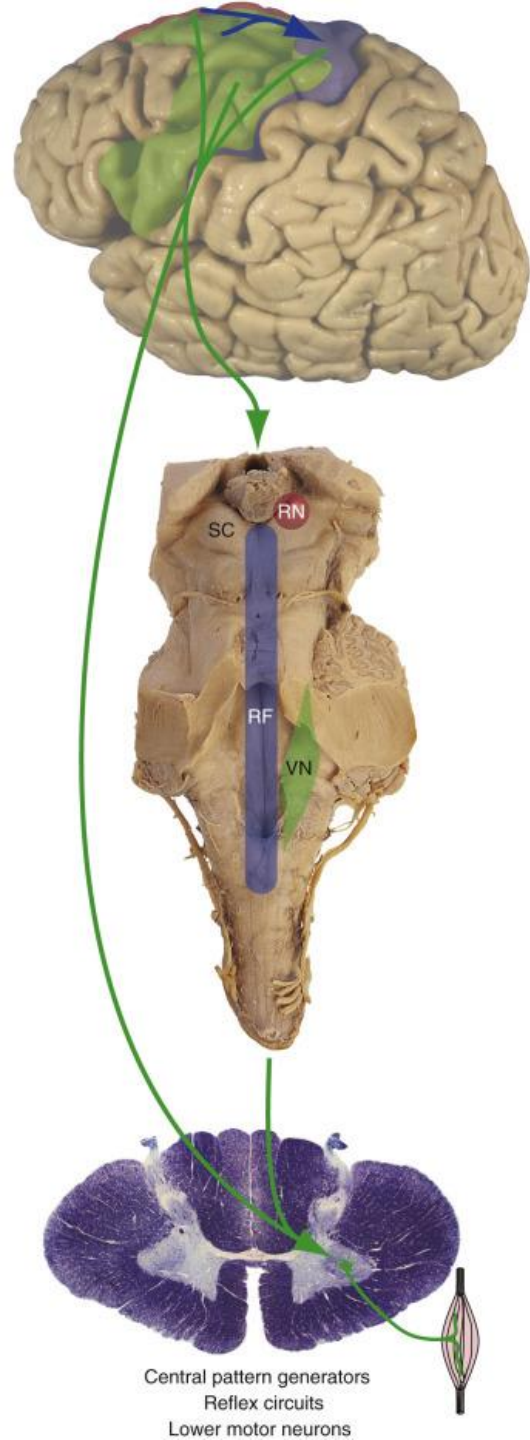


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Area 8: superior and middle frontal gyri anterior to premotor cortex and supplemental motor area
Receives inputs from visual cortical areas (contralateral hemifield), send axons to reticular formation.
Activity causes eye movements to the contralateral hemifield!
Lesions cause eyes to tonically deviate to same side, and deficits towards contralateral hemifield.



Brainstem nuclei for limb & axial control

Red nucleus – midbrain

Reticular formation – pons and medulla

Vestibular nucleus – medulla

Primarily control proximal and axial musculature, i.e. approximate reverse of cortical homunculus.

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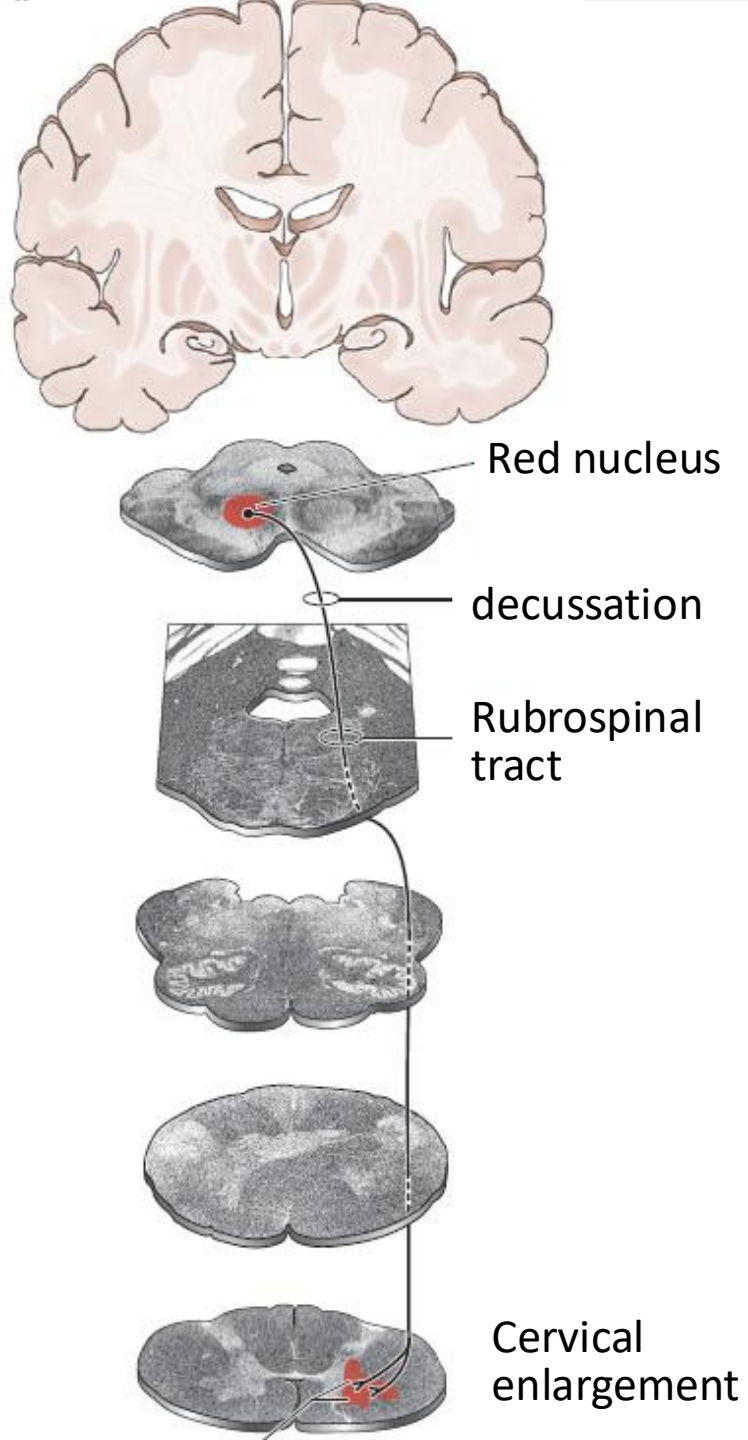
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Brainstem tracts: Rubrospinal tract

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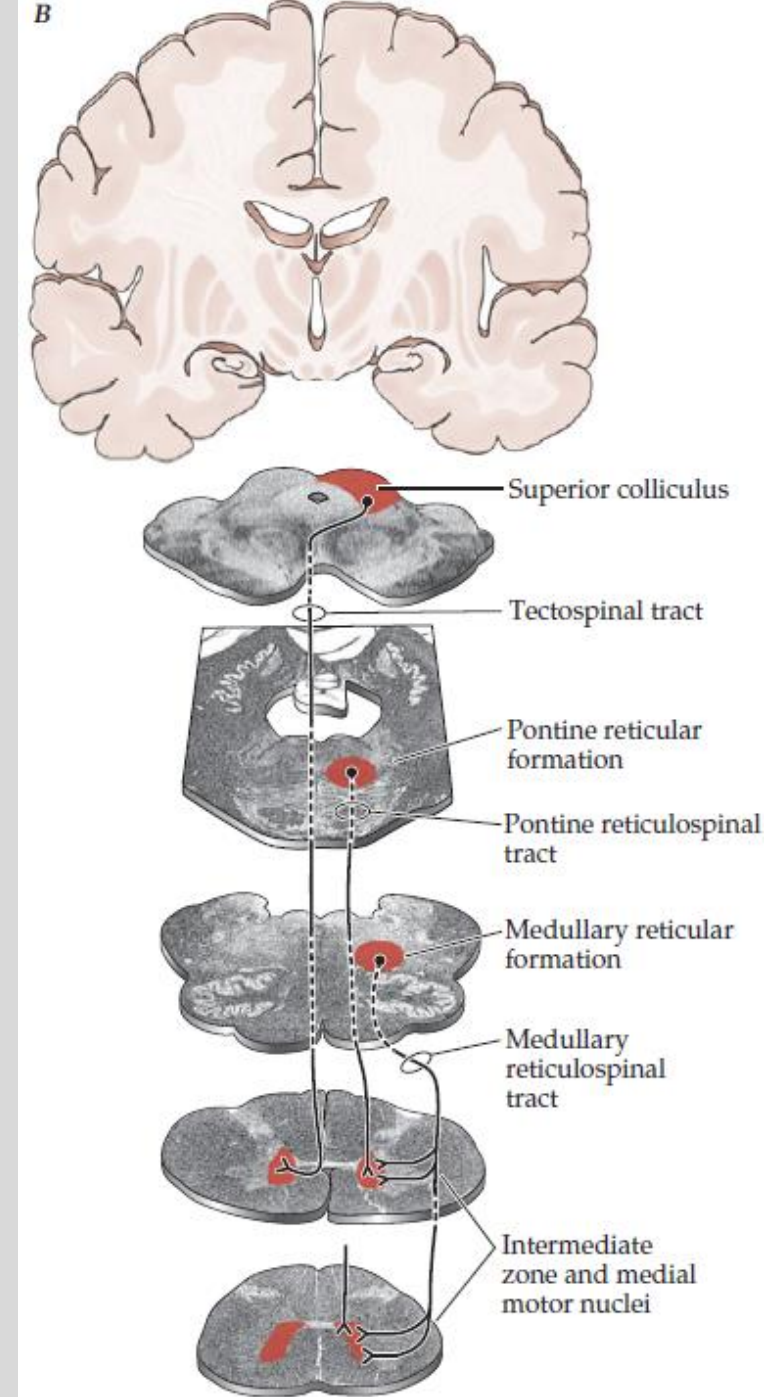
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- In midbrain
- Contralateral upper limbs
- Very small tract in humans, but large in monkeys (typical animal model)
- Red nucleus is very large in humans and has strong linkage to cerebellum

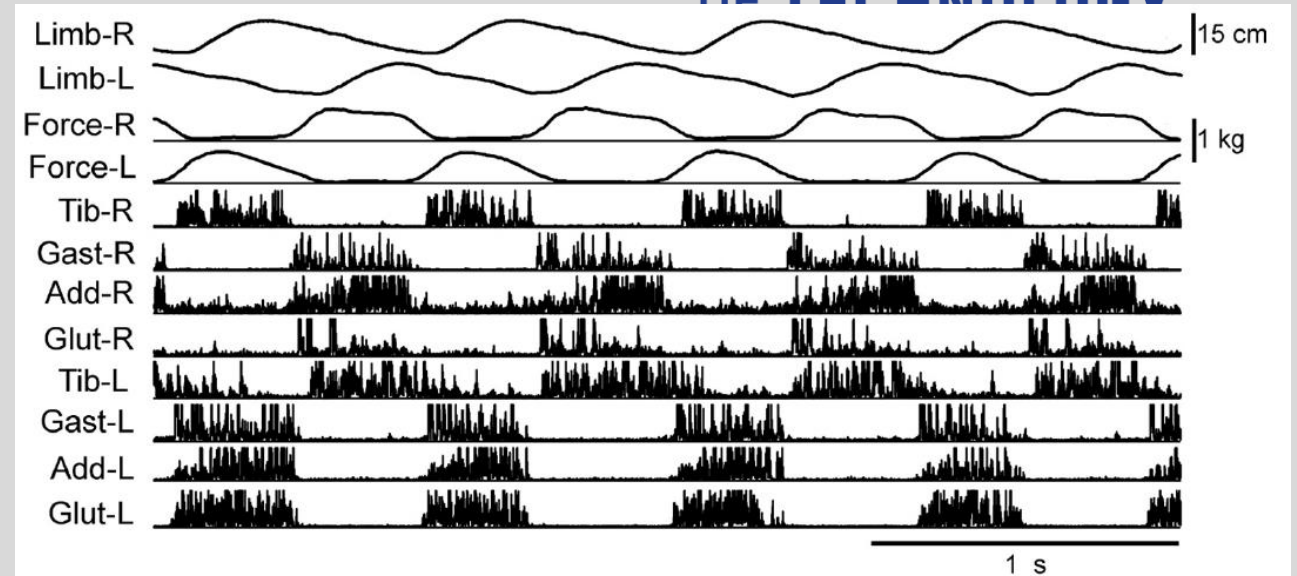
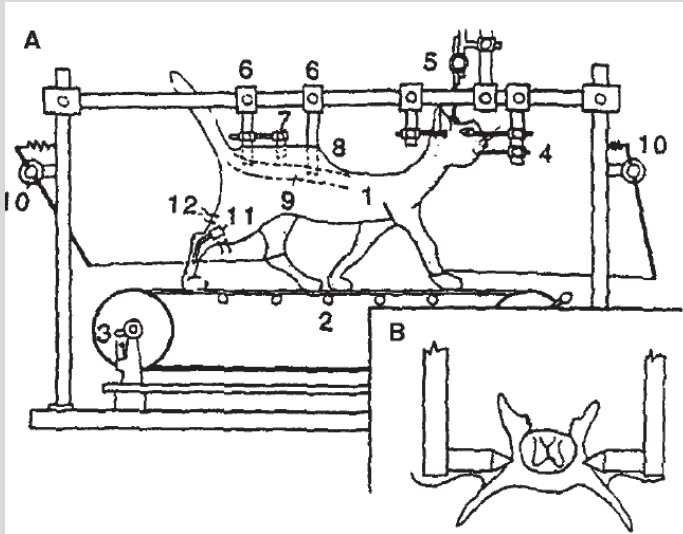
Brainstem tracts: Reticulospinal tract

- Nuclei in pons & medulla give rise to medial & lateral tracts
- Quite large but understudied (technically difficult)
- Often bilateral action of single neurons
- Receives commands from reticular formation and cerebral cortex



Brainstem tracts: Reticulospinal tract

MLR
(mesecephalic
locomotor
region)



- Stimulation leads to forward location Musienko (2012) Spinal and supraspinal control of the direction of stepping during locomotion

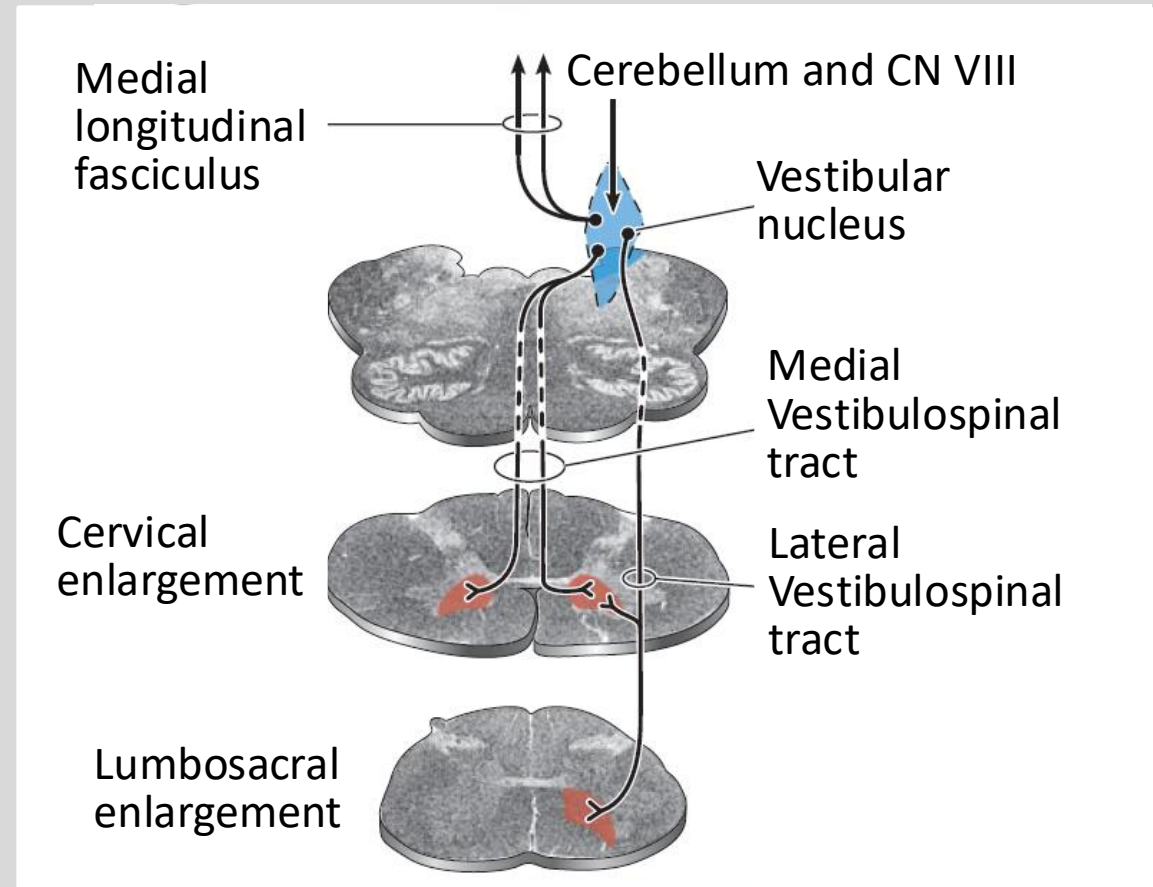
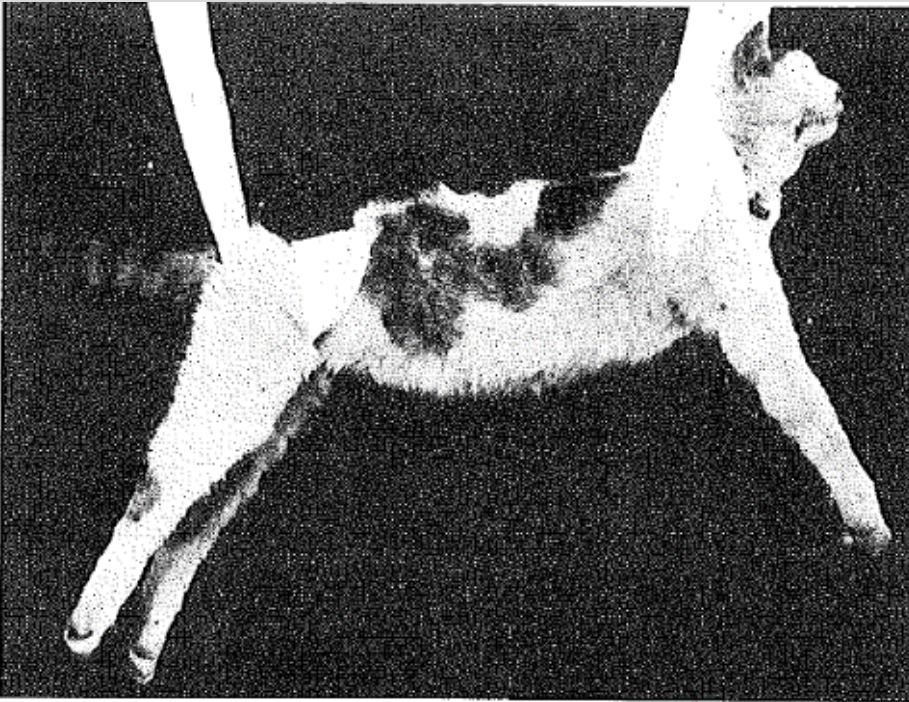
Asymmetrical tonic reflex – elbow flexion when head turns away, other arm extends. Present in infants, re-expressed with cortical stroke

Startle reflex - protective shoulder shrug and elbow flexion elicited by loud unexpected sound



Brainstem tracts: Vestibulospinal tract

- Medial vestibulospinal tract controls head and neck
- Lateral vestibulospinal tract has strong extensor activation for limb and axial muscles, e.g. activation will extend ipsilateral legs
- “Anti-gravity” function



Severe brain injury reveals overexpression of these brainstem systems

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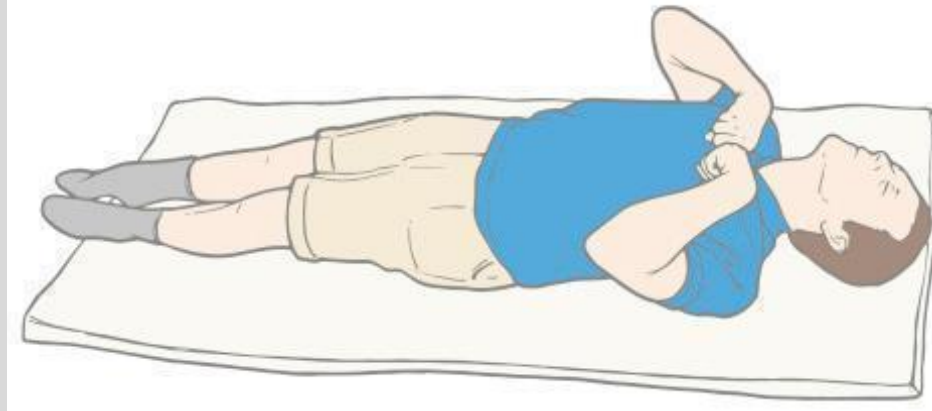
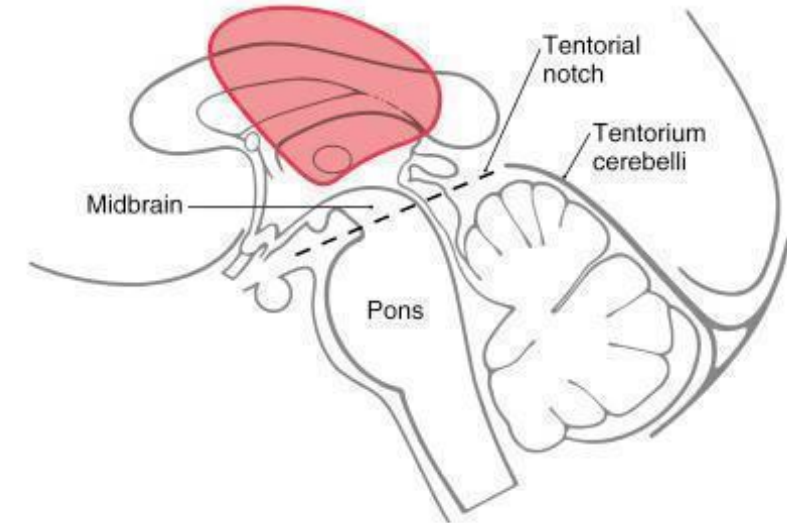
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Lesion **above** red nucleus (above superior colliculus)
➡ hyper-activity of reticular system, red nucleus,
& vestibular nuclei

“Decorticate posture”

- LE extended
- UE flexed

“Mummy pose”



Severe brain injury reveals overexpression of these brainstem systems

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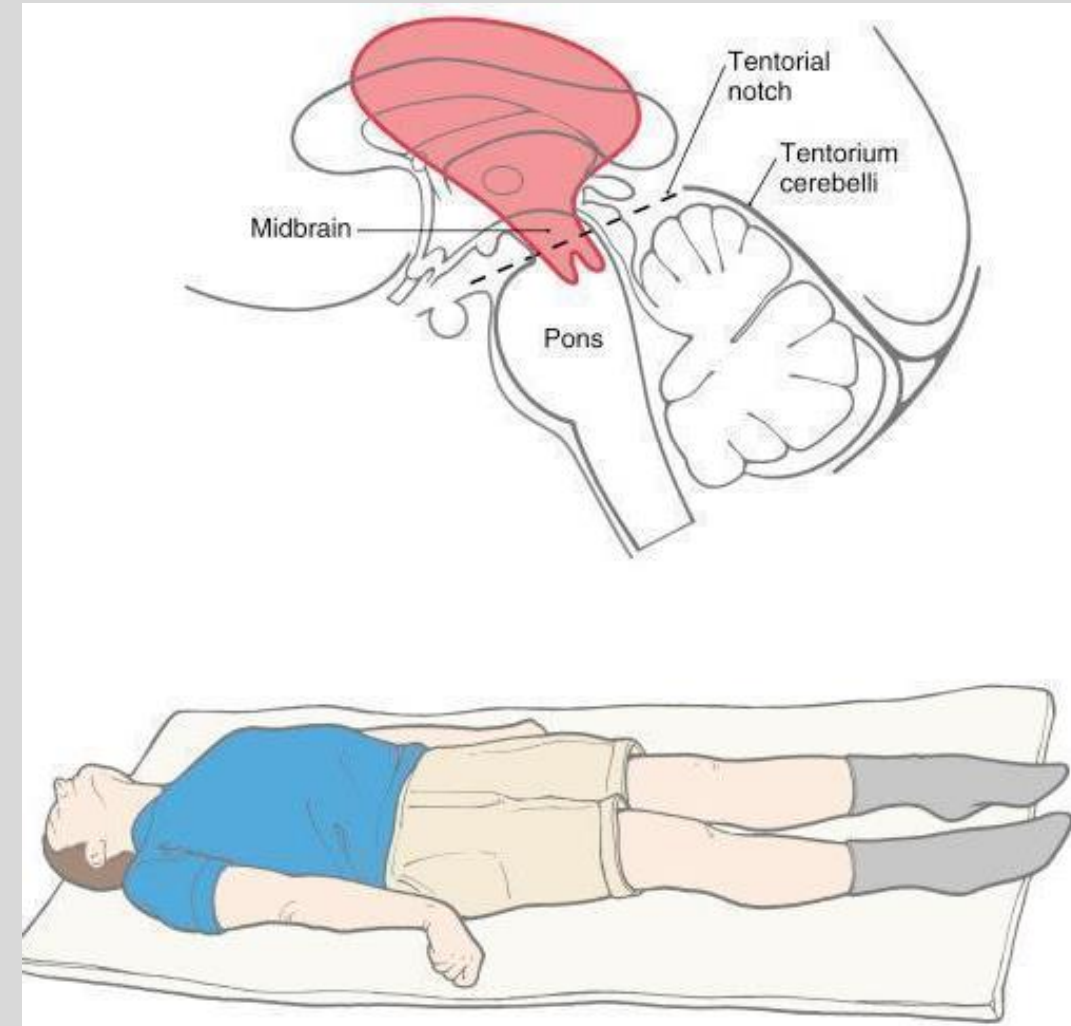
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Lesion **below** red nucleus (below superior colliculus)
➔ hyper-activity of reticulospinal and vestibular nuclei

“Decerebrate posture”

- LE extended
- UE extended

Even more severe



“Locked-In” syndrome

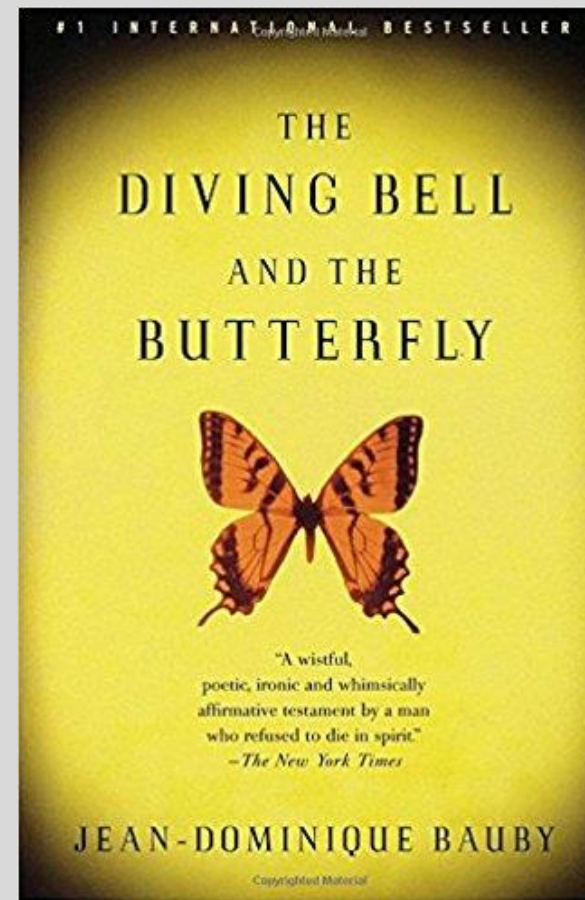


- Pontine lesion interrupts corticospinal & corticobulbar tracts
- Lose all voluntary control, except some eye movements and blinks
- Remain conscious and aware, normal sleep-cycle but “locked in” their body.
- Difficult to determine missed cases, misdiagnosed as persistent vegetative state

Improved technology to allow faster interactions

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Summary

- The corticospinal tract arises from the precentral gyrus (mostly M1) and rostral adjacent cortex (PMC and SMA) and projects to the contralateral spinal cord.
- The corticobulbar tract arises from the same areas and projects bilaterally to the cranial nerve nuclei, excepting the facial nucleus.
- Upper motor neuron damage can be expressed as weakness and spastic paralysis whereas lower motor neuron damage is expressed flaccid paralysis.
- Different descending brainstem tracts have different projection patterns.
- Damage above and below the red nucleus leads to decorticate and decerebrate rigidity, respectively.

Feedback is appreciated!

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

